

HOUSTON EXECUTIVE, TX, USA

WMO: 720617

Lat: 29.800N

Lon: 95.900W

Elev: 51

StdP: 100.71

Time Zone: -6.00 (NAC)

Period: 10-19

WBAN: 00208

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.3	(c) 0.0	(d) -8.7	(e) 1.8	(f) 3.3	(g) -6.1	(h) 2.3	(i) 6.9	(j) 11.5	(k) 14.7	(l) 10.7	(m) 13.5	(n) 2.4	(o) 330	(p) 0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.2	37.2	24.5	35.3	24.8	34.8	24.9	28.1	29.2	27.5	29.1	27.0	29.7	3.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 27.9	(b) 24.1	(c) 28.3	(d) 27.3	(e) 23.3	(f) 28.0	(g) 26.3	(h) 21.9	(i) 27.6	(j) 90.8	(k) 28.4	(l) 87.0	(m) 28.7	(n) 85.1	(o) 30.1	(p) 34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.0	(b) 8.7	(c) 7.9	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 10.0	(b) 8.7	(c) 7.9	(d) WB	(e) -5.9	(f) 28.1	(g) 2.7	(h) 1.0	(i) -7.8	(j) 28.8	(k) -9.4	(l) 29.4	(m) -10.9	(n) 30.0	(o) -12.8	(p) 30.7	(q) 30.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	11.2	13.8	17.1	20.7	24.4	27.8	28.8	29.2	26.7	21.7	16.0	12.8
	DBStd	7.50	5.43	5.84	4.87	3.68	3.14	1.64	1.31	1.70	2.27	4.07	5.66	5.50
	HDD10.0	129	50	27	7	0	0	0	0	0	0	1	12	33
	HDD18.3	790	229	146	81	19	3	0	0	0	0	16	111	186
	CDD10.0	4107	87	134	228	322	447	533	581	597	501	363	192	122
	CDD18.3	1724	7	19	43	91	192	283	323	338	251	121	40	16
	CDH23.3	17489	21	90	236	669	1725	3053	3712	4048	2447	1144	267	76
	CDH26.7	7907	1	11	34	156	637	1463	1871	2169	1103	406	52	5
Wind	WSAvg	3.3	3.6	4.0	4.0	4.1	3.9	2.9	2.6	2.6	2.5	2.9	3.4	3.6
Precipitation	PrecAvg	1224	87	73	89	105	122	113	91	100	108	124	105	91
	PrecMax	1698	235	225	212	300	311	326	297	236	320	308	365	244
	PrecMin	415	5	16	10	3	1	3	7	3	26	0	5	22
	PrecStd	322	55	52	54	80	82	74	78	61	69	99	79	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	28.0	29.0	32.2	34.1	37.4	37.3	38.8	37.1	32.9	30.0	27.4
		MCWB	19.3	20.6	21.2	21.2	23.5	23.9	24.5	24.1	23.2	24.0	23.8	21.6
	2%	DB	23.6	26.1	27.5	29.7	32.6	35.2	35.9	37.4	35.0	32.3	27.9	25.9
		MCWB	18.6	20.0	20.6	21.9	23.9	24.2	24.8	24.8	23.8	24.0	22.0	20.8
	5%	DB	22.2	24.0	26.1	28.0	32.0	34.0	34.9	36.2	33.6	30.2	27.0	23.7
		MCWB	18.7	19.2	20.0	21.4	23.8	25.0	25.2	25.1	24.3	23.3	21.5	20.5
	10%	DB	20.0	22.4	24.1	27.4	30.0	32.8	33.8	34.9	32.4	28.9	24.7	22.4
		MCWB	17.5	19.0	19.3	20.8	23.6	25.3	25.4	25.2	24.6	22.2	20.8	19.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.4	22.4	23.0	25.2	28.8	29.1	29.0	28.8	31.3	27.0	25.5	23.7
		MCDB	23.1	25.0	25.5	27.9	28.8	29.3	29.5	29.8	31.3	28.4	27.4	25.0
	2%	WB	20.7	21.7	22.3	24.0	25.9	27.5	27.7	27.8	27.8	25.6	24.1	22.6
		MCDB	22.1	24.3	25.2	27.0	29.2	29.4	29.1	29.4	28.6	28.6	26.1	24.1
	5%	WB	19.5	21.0	21.6	23.3	25.3	26.9	27.2	27.3	26.8	25.0	23.1	21.7
		MCDB	21.5	23.0	24.5	25.9	28.4	29.8	29.6	29.6	29.4	28.2	24.9	23.6
	10%	WB	17.9	20.0	20.9	22.7	24.8	26.2	26.5	26.7	26.0	24.2	22.2	20.3
		MCDB	18.8	21.8	23.2	25.4	27.9	30.0	30.8	30.6	29.3	27.4	24.1	21.8
Mean Daily Temperature Range	5% DB	MDBR	11.5	10.7	11.3	11.1	10.3	10.1	10.4	11.2	10.6	12.6	12.0	11.0
		MCDBR	12.1	11.7	11.3	11.5	10.9	11.6	11.7	13.1	12.2	12.3	11.8	12.2
		MCWBR	8.1	6.7	5.8	5.5	3.5	3.6	3.6	3.8	4.7	5.6	6.6	8.6
	5% WB	MCDBR	10.2	9.6	10.0	9.4	9.0	9.6	9.9	11.4	9.7	10.7	9.9	11.2
		MCWBR	8.0	6.8	6.7	5.8	4.2	4.4	4.0	4.0	5.3	5.8	6.8	9.5
Clear-Sky Solar Irradiance	taub	0.339	0.356	0.373	0.433	0.480	0.463	0.490	0.479	0.448	0.380	0.373	0.343	
	taud	2.483	2.451	2.394	2.232	2.139	2.242	2.186	2.218	2.269	2.444	2.407	2.485	
	Ebn at Noon	894	909	912	864	822	832	807	814	828	871	846	871	
	Edh at Noon	92	103	116	141	156	140	148	142	130	103	98	88	
All-Sky Solar Radiation	RadAvg	2.87	3.24	4.40	5.28	5.84	6.19	6.15	5.86	5.02	4.30	3.23	2.55	
	RadStd	0.28	0.58	0.38	0.40	0.43	0.48	0.55	0.42	0.50	0.52	0.34	0.31	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105	+93

Nomenclature: See separate page